LESSON 2.11

Add and Subtract Decimals

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can add and subtract decimals by lining up the place values and the decimal points.

WHY THIS MATTERS The space station's main fuel tank reads 128.75 liters. A resupply shuttle delivers 46.8 liters of fuel, but the thrusters used 19.35 liters during docking. What is the fuel level after both events?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Decimal *Spanish: Decimal*	A number with a dot, like 0.5, that shows a part less than one.	3.75 means 3 ones + 7 tenths + 5 hundredths — it is between 3 and 4
Place value *Spanish: Valor posicional*	What a digit is worth based on where it sits in a number.	In 4.38: the 4 = 4 ones, the 3 = 3 tenths (0.3), the 8 = 8 hundredths (0.08)
Tenths *Spanish: Décimas*	The first spot after the decimal point. It shows parts out of 10.	0.7 = 7/10 — like 7 slices of a pizza cut into 10 equal slices
Hundredths *Spanish: Centésimas*	The second spot after the decimal point. It shows parts out of 100.	0.05 = 5/100 — like 5 pennies out of a dollar (100 pennies)
Annex zeros *Spanish: Agregar ceros*	Adding zeros at the end of a decimal so numbers line up evenly.	15.7 becomes 15.70 so you can subtract 15.70 − 8.36 with aligned place values

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

The tank has 128.75 liters. A shuttle adds 46.8 liters. I line up the decimal points and annex a zero: 46.80.

1. I add: 128.75 + 46.80 = 175.55 liters.
2. Now the thrusters use 19.35 liters, so I subtract: 175.55 − 19.35 = 156.20.
3. The final fuel level is 156.20 liters. I check with estimation: about 130 + 47 − 19 = 158, which is close, so my answer is reasonable.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's add 12.45 and 7.8. First, how do we write 7.8 with two decimal places? (7.80.)

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- When I add _____ to _____, I move _____ to the right on the number line and land on _____.

Word bank Decimal | Place value | Tenths | Hundredths | Annex zeros | decimal | place value | line up | tenths | hundredths | right | left

WATCH OUT!

A common mistake

A common mistake in Add and Subtract Decimals is lining up the last digits (the right edges) of each number instead of lining up the decimal points

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Add and Subtract Decimals

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Show 6.5 − 2.8 and 3.25 + 1.5 on the number line



STEPS TO TRY

1. Each label shows an operation — work out the value first, then find that exact spot between the whole numbers.
2. For 6.5 − 2.8: take away 2 first to reach 4.5, then remove the extra 0.8. For 3.25 + 1.5: add 1 first, then 0.5 more.
3. Start with 6.5 − 2 = 4.5, then count back 8 tenths from 4.5 — notice you cross below 4. Place that mark, then build the sum the same way.

SHOW YOUR THINKING

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 14.6 + 3.85?

(A) 17.45
(B) 18.35
(C) 18.45
(D) 18.55

STEPS TO TRY

1. Notice the mismatch: 14.6 has one decimal place but 3.85 has two. Lining up the POINTS is the whole trick here.
2. Write 14.6 as 14.60 so both numbers show hundredths, stack the decimal points, and add column by column from the right.
3. Set up 14.60 + 3.85 with the points stacked. Add the hundredths (0 + 5) first, then the tenths (6 + 8) — that column carries.

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

3 **Circle the letter of the best answer. Show how you know.**

What is 9.05 + 4.7?

(A) 13.12
(B) 13.57
(C) 13.75
(D) 14.75

STEPS TO TRY

1. Watch the middle zero in 9.05 and the missing hundredths in 4.7 — align the decimal POINTS, never the last digits.
2. Write 4.7 as 4.70, stack the two numbers with the points lined up, then add each column starting from the right.
3. Stack 9.05 over 4.70. Add hundredths (5 + 0), then tenths (0 + 7), then the ones — watch that last sum spill into a tens place.

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

PART 3 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

What is 20.3 − 7.48?

(A) 12.18

(B) 12.82

(C) 12.92

(D) 13.82

STEPS TO TRY

1. 20.3 stops at the tenths but 7.48 has hundredths — you can't subtract from an empty column until you annex a zero.
2. Rewrite the problem as 20.30 − 7.48, stack the decimal points, and subtract right to left, borrowing when the top digit is smaller.
3. Start at the hundredths: 0 − 8 doesn't work, so borrow from the tenths to make it 10 − 8. Keep borrowing leftward as needed.

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

EXPLAIN YOUR THINKING

How does a number line help you see what happens when you add or subtract decimals?

SENTENCE STARTER When I add ___ to ___, I move ___ to the right on the number line and land on ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help